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Fast Task Adaptation Techniques in Machine Learning and Adaptive AI Systems.

Fast Task Adaptation has emerged as a critical research area in Artificial Intelligence (AI), Machine Learning (ML), and Reinforcement Learning (RL) due to the increasing demand for intelligent systems capable of rapidly adapting to dynamic and uncertain environments with minimal retraining. Traditional machine learning approaches often require large scale labeled datasets, extensive computational resources, and repeated retraining processes, limiting their effectiveness in real time and evolving applications. This survey presents a comprehensive review of modern Fast Task Adaptation techniques with a primary focus on Meta-Learning, Meta-Reinforcement Learning, Hierarchical Reinforcement Learning, Federated Learning, Curriculum Learning, Continual Learning, and adaptive intelligent architectures. The paper analyzes the principles, architectures, optimization methods, and evolutionary trends of adaptive learning systems that enable efficient knowledge transfer, contextual adaptation, and rapid decision-making across diverse domains. Furthermore, the survey examines the role of advanced technologies such as Graph Neural Networks (GNNs), Transformer architectures, attention mechanisms, distributed adaptive intelligence, and self-supervised learning in improving scalability, generalization, and sample efficiency. Various real-world applications including robotics, cyber security, healthcare, intelligent tutoring systems, wireless communication, smart energy management, and autonomous multi-agent systems are discussed to demonstrate the practical significance of Fast Task Adaptation. The survey also highlights major research challenges such as computational complexity, catastrophic forgetting, training instability, scalability limitations, privacy concerns, and real-time deployment issues. Finally, future research directions toward scalable, autonomous, explainable, and human-centric adaptive AI systems are presented to support the development of next-generation intelligent systems and Artificial General Intelligence (AGI).

Uploaded by



Shantveer Kesti

shantveerkesti045@gmail.com

Coauthors



Harish Kundar

aietaiml08@aiet.com



Shantveer Kesti

shantveerkesti045@gmail.com



Madhusudhan G

madhugreddy30@gmail.com



Pavankumar A

pavankumarpk88@gmail.com



Kutakole Gauri Mahadev

gaurikutakole2003@gmail.com



Pawade Alekha Pratap

pawadealekha@gmail.com

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Assigned peer-reviewers

Rachmad Vidya Wicaksana Putra

rachmad.p****@nyu.edu

<https://www.webofscience.com/wos/author/record/AAM-7149-2020>

Researcher ID: AAM-7149-2020

Orcid ID: 0000-0001-8597-4530

New York Univ NYU Abu Dhabi

U ARAB EMIRATES

Jianguo Ding

jianguo.****@bth.se

<https://www.webofscience.com/wos/author/record/B-5748-2008>

Researcher ID: B-5748-2008

Orcid ID: 0000-0002-8927-0968

Blekinge Institute Technology

SWEDEN

ZIP: S-37179

Maria Trigka

mtr****@uniwa.gr

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